

Cow's-milk-free diet as a therapeutic option in childhood chronic constipation.

OBJECTIVES: It has been reported that a number of children with constipation respond to a diet free of cow's-milk (CM) proteins, although evidence is lacking to support an immunoglobulin E-mediated mechanism. **PATIENTS AND METHODS:** We performed an open-label crossover study comparing CM and rice milk in 69 children who fulfilled Rome III criteria for chronic constipation. Clinical, physical, and immunologic parameters of patients who responded (R) and who did not respond (NR) to a CM-free diet were compared. **RESULTS:** Thirty-five of the 69 children (51%) improved during the first CM-free diet phase, 8 of these did not develop constipation when CM was reintroduced, and 27 children (39%) developed constipation during the CM challenge and improved during the second CM-free diet phase (R group). Thirty-four children (49%) did not improve during the first CM-free diet phase (NR group). Bowel movements per week among R children significantly increased compared with NR children (R: 2.8-7.7 vs NR: 2.6-2.7) ($P < 0.001$). Seventy-eight percent of the children with developmental delay responded to the CM-free diet ($P = 0.007$). No significant statistical difference was found between the R and NR children in terms of fiber and milk consumption; atopic or allergic history; full-blood eosinophil count and percentage, and lymphocyte populations; immunoglobulins, immunoglobulin (Ig)G subclasses, total IgE; and serum-specific immunoglobulin E for CM proteins. **CONCLUSIONS:** A clear association between CM consumption and constipation has been found in more than one third of children. However, analytical parameters do not demonstrate an immunoglobulin E-mediated immunologic mechanism.